Oliver Dong, Chen

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EDUCATION

PhD Department of Orthopaedics and Traumatology, The University of Hong Kong | Feb 2023 - Expected Dec 2026

- **Thesis:** "Development of A Gait Video-Based Screening Method for Adolescent Idiopathic Scoliosis Using Multimodal Deep Learning".
- The main jobs included: 1) Raw data collection at clinic. 2) Multimodal data construction and curation. 3) Representation learning. 4) Pretraining + supervised fine-tuning. 5) Model interpretation with clinical priors.

MSc Human and biological robotics (Bioengineering), Imperial College London | Merit | 2020 - 2021

- **Thesis: Super-Resolution Bubble Localization (Python/MATLAB):** Optimized encoder-decoder CNN architectures, improving localization precision in dense ultrasound scenarios.
- **Human-Centered Assistive Device (Python/Arduino):** Developed a computer vision system integrating pose estimation and wearable sensors for stroke rehabilitation, enhancing patient engagement via real-time music feedback.
- **Reinforcement Learning Projects:** Applied TD learning, Markov Decision Processes, and Monte Carlo algorithms to solve dynamic control problems.

BEng Electrical Engineering, University of Liverpool & Xi'an Jiaotong-Liverpool University | 1st Class Honour | 2016 - 2020

- **Thesis:** Developed an NLP-driven ML framework to analyze metal-organic framework (MOF) synthesis parameters, enabling 80% faster data extraction from scientific literature.

Research Experiences

- **Research Intern (part-time)** | Bone's Technology | Mar 2026 - July 2026
  - Lead sub-group muscle segmentation on CT/MRI for sarcopenia diagnosis support
  - Built data construction pipelines and clinical-prior-based pretraining with supervised fine-tuning
- **Research Intern** | Huawei Technologies (China) | Jun 2025 - Dec 2025

  \- Project Leader: Token Entropy Regularization for Multi-modal Antenna Affiliation Identification. I designed the framework of this task and achieved this project in five months with supports from two data engineers. arXiv: https://arxiv.org/abs/2601.21280.

  \- Others: Fine-tune large visual & language Model *(CVPR 2026 - Spacemind)*. arXiv:2511.23075.
- **Student Researcher** | The University of Hong Kong - Shenzhen Hospital | Aug 2023 - Apr 2025

  \- Project Participant: Integrated intelligent robots for research wards. I drafted the gait monitoring part of research proposal, literature review, and methodology sections. The video-based human behaviour analysis is based on my PhD thesis applied in surgical ward application. *(Funded)*
- **Research Assistant** | University of Hong Kong | Nov 2021 - Oct 2022

  \- Project Leader: Deep Learning-Based Scoliosis Detection. Designed a CNN-based architecture for analyzing holistic motion, achieving 85% accuracy in a pilot study of early scoliosis screening. Collaborated with clinicians to validate results on a dataset of 500+ patient videos. *(42nd HKOA)*

Publications

- **Chen, D.**, Li, R., Zhang, X., Xu, J., Zhao, R., Zhang, Z., Li, L. and Wei, Z., 2026. Token Entropy Regularization for Multi-modal Antenna Affiliation Identification. arXiv preprint arXiv:2601.21280.
- Zhao, R., Zhang, Z., Xu, J., Chang, J., **Chen, D.**, Li, L., Sun, W. and Wei, Z., 2025. SpaceMind: Camera-Guided Modality Fusion for Spatial Reasoning in Vision-Language Models. arXiv:2511.23075. [**CVPR 2026**]

- **Dong Chen**; Huili Peng; Kenneth MC Cheung."Revolutionizing Digital Transformation in Adolescent Idiopathic Scoliosis Screening: Ai-Assisted Holistic Motion Analysis Via a Smartphone Camera for Scalable, Cost-Effective Screening", Podium Presentation at the 60th Scoliosis Research Society (SRS) Annual Meeting 2025, Charlotte, North Carolina, USA.
- **Chen, D.**, Wei, Y., He, Z., Kuang, G.M., Ye, C., An, M., Peng, H., Hu, Y., Tao, H. and Cheung, K., 2025. Diagnosing Hallucination Risk in AI Surgical Decision-Support: A Sequential Framework for Sequential Validation. arXiv preprint arXiv:2511.00588.
- **Chen, D.,** Peng, H., Hu, Y. and Cheung, K., 2025. Selecting Optimal Camera Views for Gait Analysis: A Multi-Metric Assessment of 2D Projections. arXiv preprint arXiv:2509.17805.
- **Dong Chen;** Yong Hu; Kenneth Man Chee Cheung, 'A Deep Learning-based Motion Video Analysis for Scoliosis Screening', The Hong Kong Orthopaedic Association 42nd Annual Congress, November, 2022.

Others and under review:

- **Dong Chen**, Kenneth MC, Cheung, Huili Peng. CN 2024116816723 — ST-iTransformer training method for disease prediction (substantive examination, Nov 2024)
- **Dong Chen**, Kenneth MC, Cheung, Huili Peng. CN 2024116901637 — Camera angle-adaptive normalization for scoliosis screening (substantive examination, Nov 2024)
- **Dong Chen**, Kenneth MC, Cheung, Huili Peng. CN 2024116896569 — Gait-direction-based preprocessing for scoliosis screening (substantive examination, Nov 2024)
- **Dong Chen**, Kenneth MC, Cheung, Huili Peng. CN 2024116897415 — Displacement-based preprocessing for scoliosis screening (substantive examination, Nov 2024)
- **Dong Chen**, Kenneth MC, Cheung, Huili Peng. CN 2024116795074 — Top-down/bottom-up combined preprocessing for scoliosis screening (substantive examination, Nov 2024)
- **Dong Chen**, Kenneth MC, Cheung, Huili Peng. CN 2024116932211 — Medical data management device / electronic device / robot (substantive examination, Nov 2024)
- **Dong Chen**, Kenneth MC, Cheung, Huili Peng. CN 2024116801253 — AI-based scoliosis screening system (substantive examination, Nov 2024)
- **Dong Chen**, Kenneth MC, Cheung, Huili Peng. WO/CN25107777 — Disease prediction model PCT application (priority Nov 2024; filed Jul 2025)
- **Dong Chen**, Kenneth MC, Cheung, Huili Peng. ScoliDetect System V1.0 — CN Software Copyright [2025SR0359659]
- **Dong Chen**, Kenneth MC. Know-how: Instructions for Environment Setting and Machine Learning Operations — HKU TTO, 2022

Exhibitions & Rewards

- Silver Medal, 49th International Exhibition of Inventions of Geneva (2025). Awarded for the project "ScoliDetect: An AI-Powered, Video-Based Gait Analysis Model for Non-Invasive Scoliosis Screening" (as part of the HKU team).
- Key Exhibit of HKU, presented during the VIP tour at "InnoCarnival 2025" ;
- Key Exhibit of HKU - Shenzhen Hospital at the 2024 BrightMD Fair China, Shenzhen.

Certifications

- Quantization in Depth (DeepLearning.AI) | 2024
- Machine Learning with Python (IBM/Coursera) | 2021
- Natural Language Processing in TensorFlow (DeepLearning.AI/Coursera) | 2021

Key skills

- Programming: Python (TensorFlow, PyTorch), C/C++, MATLAB
- Tools: OpenCV, Huggingface, Git, Vibe coding (Cursor)